Combinatorics 1

Binomial Coefficients → ways to choose

- Number of ways to choose K items from N items

${}^{n}C_{k}$

same thing

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

2, 1

1, 2

5

1 2 3 4 5

5 elements

3 elements

$$\frac{5 \cdot 4 \cdot 3}{k!}$$ → number of ways to pick when order matters.

1, 2, 3 ⟷ (diff) 3, 2, 1

1, 2, 3, 4, 5

1, 2, 3

3, 2, 1

1, 3, 2

2, 1, 3

2, 3, 1

3, 1, 2

k = 3

k!

^nC_r

$\Rightarrow n \cdot (n-1) \cdot (n-2) \cdots (n-r+1)$

$$\frac{n!}{(n-r)!} \cdot \frac{1}{r!} = {}^{n}C_{r}$$

$(n-r)!$

$\underbrace{1 \cdot 2 \cdots (n-r)}\,(n-r+1) \cdots n$

$\underbrace{1 \cdots (n-r)}$

${}^{n}C_{r}$

$n \le 10^6$

$r \le n$

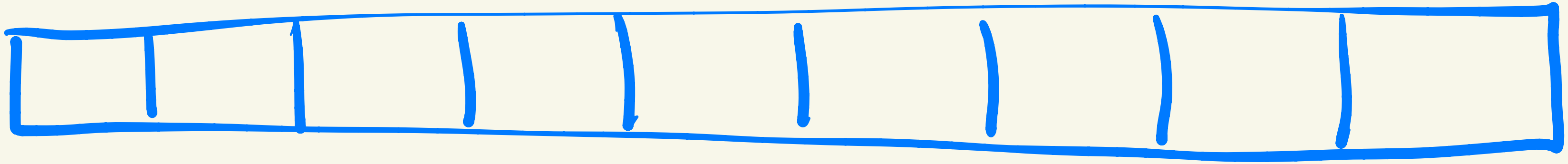
$\underbrace{\phantom{{}^{n}C_{r}}}$ → Really large % P

$P = 10^9 + 7$

$$\underbrace{{}^{n}C_{r} \ \% \ P}$$

$\underbrace{q} \le 10^5$ $\underline{r \le n \le 10^6}$

$n \quad r \to {}^{n}C_{r} \ \% \ \underbrace{10^9+7}$



$i \to \underline{i!}$ $1 \to 10^6$

$\to \boxed{\left[\frac{1}{i!}\right] - (i!)^{-1}}$

$$\frac{n!}{(r!\,(n-r)!)} \rightarrow \underbrace{n! \cdot (r!)^{-1} \cdot (n-r!)^{-1}}$$

$O(1)$

$$\frac{1}{n!} \,\%\, M = (n!)^{M-2}$$

Fermat's little theorem

$i! \,\%\, M$ $\quad$ $fact[0] = 1$ $\quad$ $n! = n \cdot (n-1)!$

```
for(i=1, i ≤ 10^6, i++)
    fact[i] = (fact(i-1) · i) % MOD
```

```
for (i=1, i ≤ 10^6, i++)
    ifact[i] = pow(fact[i], M-2)
```

$$ifact[i] = \underbrace{pow(fact[i], M-2)}_{\text{Binary Exponen}}$$

$O(n \log M)$

$a^b \to O(\log_2 b)$ Binary Exp

$\to O(n \log M) \leftarrow$ Good ✓

$\frac{1}{n!} = (n!)^{-1}$

$((n-1)!)^{-1}$

$\boxed{\frac{1}{(n-1)!}} = \frac{n}{n!}$

```
for (i = 1; i <= n; i++)
    fact(i) = (i * fact(i-1)) % M

x [invfact(n) = binexpo(fact(n), M-2)]

for (i = n-1; i >= 1; i--)
    invfact(i) = (invfact(i+1) * (i+1)) % M
```

$O(n)$

$$\frac{1}{i} = \frac{i+1}{(i+1)!} = \frac{\cancel{i+1}}{\cancel{(i+1)}\,(i)!}$$

Precomp → $O(n)$

Each query → $\underline{O(1)}$

$\underbrace{{}^{n}C_{r}}$ % MOD $= ((\underline{fact(n) * ifact(n-r)}$ % $\underline{MOD})$

$* \underline{iFact(r)})$ % $\underline{MOD}$

Requirements to calculate C(n, r)

- Mod Inverse calculation
- Precomputation of Factorials
- Precomputation of Inverse Factorials

Precomputation Trick O(N)

- Can we optimise the O(NlogM) precomputation time to just O(N)?

    - inverse_fact[n - 1] = inverse_fact[n] * n
    - we can first calculate inverse_fact[n] in O(logM) time then find out all the inverse factorials from n - 1 to 1 in O(1) using the above formula

Efficient Implementation

```
ll combination1(ll n, ll r, ll m, vector<ll>& fact, vector<ll>& ifact){
    return mod_mul(fact[n], mod_mul(ifact[r], ifact[n - r], m), m);
}
void solve(){
    int n = 10;
    vector<ll> fact(n + 1);
    vector<ll> ifact(n + 1);
    fact[0] = 1;
    for(int i = 1; i <= n; i++){
        fact[i] = mod_mul(fact[i - 1], i, MOD);
        ifact[i] = mminvprime(fact[i], MOD);
    }
    ifact[n] = mminvprime(fact[n], MOD);
    for(int i = n - 1; i >= 0; i--){
        ifact[i] = mod_mul(ifact[i + 1], i + 1, MOD);
    }

    cout << combination1(8, 6, MOD, fact, ifact) << endl;
}
```

inverse factorial

$\log_2 (m)$

```
int mod_mul(a, b, MOD){
    (a%MOD * b%MOD) %MOD
}
```

Made with Goodnotes

A different way to look at C(n, r)

C(n, r) = (n * (n - 1) * (n - 2) … * (n - r + 1)) / (1 * 2 * 3 … * r)

2 different use cases:

- Calculate C(n, r) many times with O(n) precomputation and O(1) calculation of each query $n, r \le 10^6$
- Calculate C(n, r) just once in O(r) time

$n \le 10^9$

$\boxed{r \le 20}$

$$\boxed{{}^{n}C_r \ \% \ MOD}$$

% MOD atmost 20 iterations

$$\underbrace{n \cdot (n-1) \cdots (n-r+1)}_{r \text{ numbers}}$$

$${}^{n}C_r = \frac{n!}{r!\,(n-r)!}$$

$O(10^9)$ → TLE

$O(r)$ $r \le 20$

Made with Goodnotes

Important Binomial Results

$$\boxed{\binom{n}{k} = \binom{n}{n-k}}$$

$^{n}C_{k} = {}^{n}C_{n-k}$

$$\boxed{\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}}$$

$$\boxed{\sum_{k=0}^{n} \binom{n}{k} = 2^n}$$

$\_\ \_\ \_\ \_\ \_\ \_\ \_\ \ n$

choose k elements $= {}^nC_k$

given n elements, discard $n-k$ elements

$${}^nC_k = {}^nC_{n-k}$$

$$^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$$

$$r = n - n + r$$

$$\underbrace{r}_{} = \underbrace{n - (n - r)}_{}$$

Math Proof

$$^{n}C_{r} = \frac{n!}{(n-(n-r))!}\;(\cancel{n}-\underline{\quad\cancel{n}}+(n-r))!$$

$$= \frac{n!}{(n-(n-r))!\,(n-r)!}$$

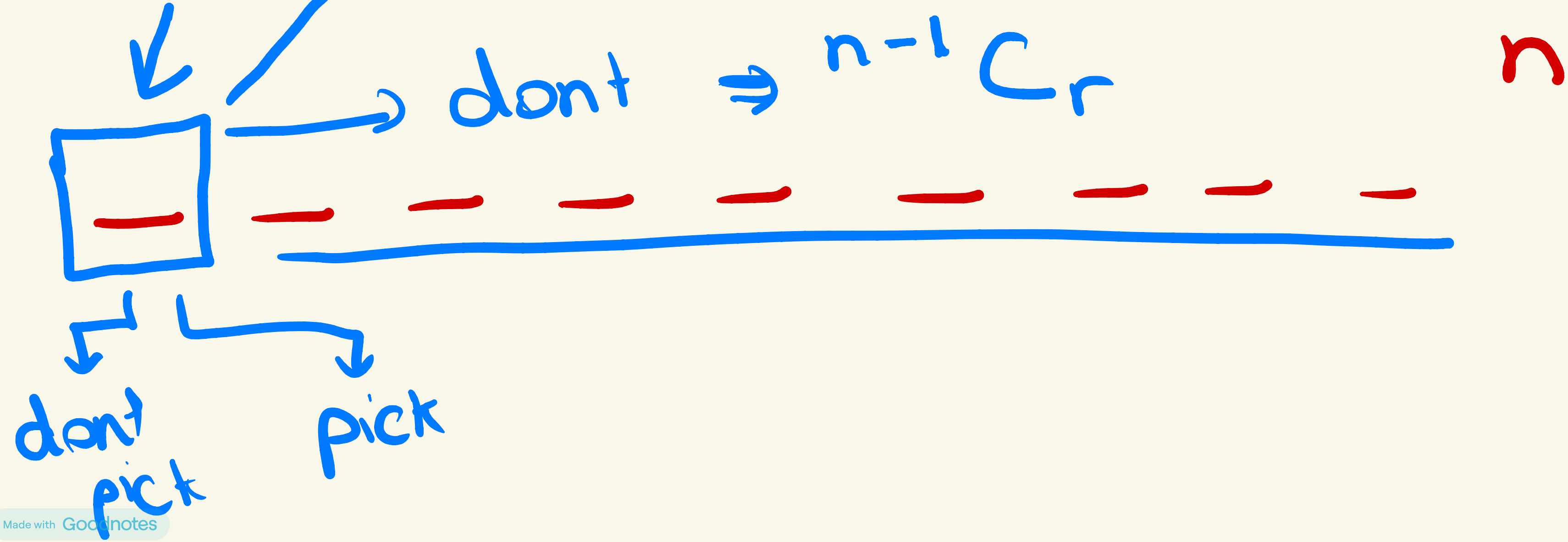
$$= {}^{n}C_{n-r}$$

$${}^{n}C_{r} = {}^{n-1}C_{r-1} + {}^{n-1}C_{r}$$

pick $= {}^{n-1}C_{r-1}$

$+$

dont $\Rightarrow {}^{n-1}C_{r}$



$$\boxed{\sum_{r=0}^{n} {}^{n}C_r = 2^n} \leftarrow \text{Formula}$$

$n = 3$

no. of subsets of size 1

3

$$\hookrightarrow \underbrace{{}^3C_0}_{\text{number of subsets of size 0}} + \overbrace{{}^3C_1}^{\text{no. of subsets of size 1}} + \underbrace{{}^3C_2}_{2} + \overbrace{{}^3C_3}^{3}$$

= Total no. of subsets

$\boxed{\text{Total number of subsets of an array of size } n}$

$$\underline{2} \times \underline{2} \times \underline{2} \times \underline{2} \times \underline{2} \times \underline{2}$$

not pick

pick

$$\rightarrow 2^n$$

Important Binomial Results

- K items always included - C(n - k, r - k)
- K items are never included - C(n - k, r)
- Value of C(n, r) is greatest when r = n/2

$\hookrightarrow {}^nC_r \rightarrow \underbrace{r = n/2}$

${}^nC_r = {}^nC_{n-r}$

${}^nC_0 \quad {}^nC_1 \quad {}^nC_2 \quad \ldots \quad {}^nC_{n-2} \quad {}^nC_{n-1} \quad {}^nC_n$

k

n choose r elements

$k \le r$

$\boxed{{}^{n-k}C_{r-k}}$

k

gone

$\underbrace{r-k}$

K people chosen on Random

$${}^{n}C_{k} \cdot {}^{n-k}C_{R-k}$$

$$\underbrace{n \text{ people}}_{r \text{ people}}$$

$$\longrightarrow \boxed{{}^{n-k}C_r}$$

Arrangement

Permutations

- Arranging distinct elements

$$\boxed{n!}$$

$\underline{n}\ \underline{n-1}\ \underline{n-2}\ \ldots\ \underline{1}$

* Arranging similar elements

$$\frac{(a_1+a_2+a_3+\ldots+a_n)!}{a_1!\cdot a_2!\cdot\ldots\cdot a_n!}$$

$a_1 \; a_2 \; b$

$a_1 \; b \; a_2$

$b \; a_1 \; a_2$

$b \; a_2 \; a_1$

$a_2 \; b \; a_1$

$a_2 \; a_1 \; b$

$3! = 6$ ways.

$a_1 \quad a_2 \quad a_3$

$\underline{3} \quad b \quad \underline{2} \quad \underline{1}$

n

$a_1 \quad a_2 \quad a_3 \ldots \quad a_k$

$$\frac{\left(\sum a_i\right)!}{(a_1!)(a_2!)\cdots(a_k!)}$$

$$\underbrace{(a_1\ a_2\ a_3)}_{3!}\ \ \times\ \ \underbrace{b_1\ b_2\ b_3}_{3!}$$

Let’s solve some problems

- Creating Strings 2 - Link
- Kth String in Dictionary → n
- Unique Paths - Link
- Arrange N different items such that K of them always come together

CSES

a a b a c

$$\frac{5!}{3!\,1!\,1!} = \frac{5!}{\boxed{3!}} = 5 \times 4 = \underbrace{20}$$

inverse % P

$$\left(\boxed{3!}\right)^{-1}$$

$$x^{M-2} \equiv x^{-1} \ \% \ M$$

$\hookrightarrow$ binary exponentation

$n = 3$ k

Kth string

1 a a a
2 a a b
3 a a c
a a d

27th a b a
28th a b b

4

a c b c c c c d . . . . c z

Number system of size 26 digits

n = 2

%26

0 a

1 a

2 a

3 a

~~26~~ b

~~27~~ b

~~28~~ b

2 c

int → binary

$\underbrace{x = k-1}$ s =

(i = 0 ; i < n ; i++) {

d = x % b 26 → 0 – a
                    1 – b

x /= b 26

s += d

}

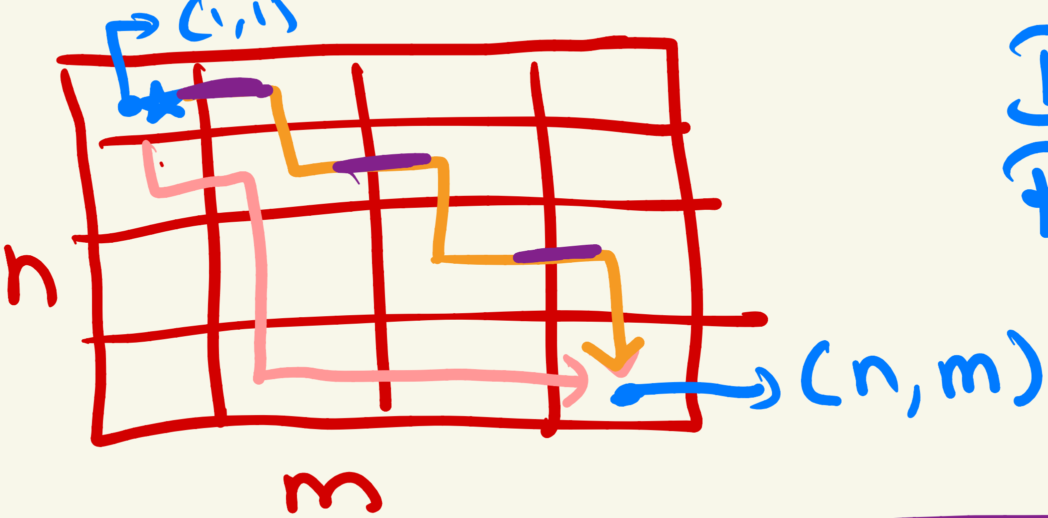
return reversed(s)

$\boxed{b}$

b = 2

b = 3

$\boxed{b = 26}$



Down
Right

DP $O(n \cdot m)$

Math $O(n+m)$

$m-1$ Right Moves
$n-1$ Down Moves
→ constant

$$\frac{(n-1+m-1)!}{(n-1)! \cdot (m-1)!}$$
→ Total number of paths.

R R R D D D → Number of arrangements of this.

R D D R D

$$\frac{n+m-2 \text{ moves}}{n-1} \rightarrow D$$

$$\boxed{\frac{{}^{n+m-2}C_{n-1}}{{}^{n+m-2}C_{m-1}}}$$

_ _ _ _ _ _ _ _

$${}^{n}C_{r} = {}^{n}C_{n-r}$$

$${}^{n+m-2}C_{n-1} = {}^{n+m-2}C_{m-1}$$

$$(n+m-2) - (n-1) = m-1$$

$\cancel{k}$ k 3

group → 1 elm
→ 3 elm

1 2 3 $\cancel{4}$ $\cancel{5}$ 6

R

+ ↳ pick

↳ dont

$$\boxed{{}^{n-k}C_{R-k} + {}^{n-k}C_{R}}$$